

Harsha Varthanan P

Materials Science Engineer — Computational Materials Science — Machine Learning Applications

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Education

Anna University, College of Engineering Guindy (CEG)

Chennai, India

Bachelor of Engineering in Materials Science and Engineering

CGPA: 8.9/10

Expected Graduation: 2028

Professional Summary

Materials Science and Engineering undergraduate with experience in advanced metal joining technologies, finite element analysis, materials characterization, machine learning, and engineering data analytics. Completed the IIM-PRIME 2026 Research Internship at IIT Indore, investigating nano-oxide activated flux systems (TiO_2 , ball-milled TiO_2 , SiO_2 , and MoO_3) for Activated TIG (A-TIG) welding, with emphasis on penetration behaviour, microstructural evolution, and mechanical properties. Served as the Research Project Team Lead for the CEG Tech Forum (K-Research 2025–26), developing and simulating a Nitinol shape-memory-alloy actuator for adaptive lunar drilling applications using COMSOL Multiphysics. Additionally, designed and modelled hybrid PZT/PVDF piezoelectric energy harvesting systems for flexible sensing applications. Interested in Computational Materials Science, Materials Informatics, Scientific Machine Learning, Smart Materials, and Advanced Manufacturing.

Experience

Indian Institute of Technology Indore

Summer Research Intern (IIM-PRIME 2026 Research Internship)

May 2026 – June 2026

- Conducted experimental research on nano-oxide-assisted metal joining by ATIG process using engineered oxide systems based on TiO_2 , ball-milled TiO_2 , SiO_2 , and MoO_3 .
- Evaluated process-performance relationships through metallographic characterization, microhardness testing, and weld geometry analysis.
- Investigated the influence of oxide particle engineering and processing conditions on penetration behaviour, microstructural evolution, and mechanical response.
- Applied data-driven analysis techniques to correlate process parameters with performance metrics.

CEG Tech Forum (K-Research 2025–26) Research Project Team Lead

Dec 2025 – May 2026

- Led development of a smart-material-driven adaptive actuation system for lunar regolith drilling applications.
- Designed a Nitinol shape-memory-alloy (SMA) actuator capable of generating passive thrust and anchoring forces in low-gravity environments.
- Performed finite element simulations using COMSOL Multiphysics and the Lagoudas SMA constitutive model to study shape recovery and electrically induced actuation behaviour of NiTi systems.
- Evaluated displacement response and strain generation during SMA phase transformation under electro-thermal loading conditions.
- Simulation results predicted approximately 700–900 N thrust force generation with 5–7 mm actuator displacement, demonstrating feasibility for lunar drilling applications.
- Presented project outcomes at TECHNOVATION 2026 and Kurukshetra 2026.

Indian Institute of Metals (IIM ATM 2025), IIT Hyderabad Research Poster Presenter

Dec 2025

- Presented a research poster titled “*Highly Sensitive Flexible Footstep Energy Harvester Sensor Using PZT Thin Film on PVDF Substrate*” at the International Conference on Advanced Materials and Critical Minerals for Energy Transition (IIM ATM 2025), IIT Hyderabad.
- Designed and simulated a hybrid piezoelectric energy harvester integrating PZT thin films and PVDF on a silicone substrate using COMSOL Multiphysics.
- Developed an interdigitated PZT–PVDF architecture to improve strain distribution, structural flexibility, and voltage generation performance under biomechanical loading conditions.
- Investigated deformation behaviour, piezoelectric response, and electric potential generation across applied pressure conditions ranging from 10 kPa to 300 kPa.

- Demonstrated higher voltage output from PZT regions while utilizing PVDF to enhance flexibility and reduce localized stress concentrations.

Projects

Machine Learning for Process-Performance Prediction in Advanced Manufacturing

- Developed machine-learning models using Python, Scikit-Learn, and XGBoost for process-performance prediction.
- Built datasets from experimental observations and published engineering literature.
- Achieved approximately $R^2 = 0.85$ using XGBoost regression.
- Developed an interactive Streamlit dashboard for process parameter exploration and prediction.

Flexible Piezoelectric Footstep Energy Harvester

- Designed and simulated flexible piezoelectric energy harvesting structures using COMSOL Multiphysics.
- Investigated electromechanical coupling and voltage amplification mechanisms in PZT/PVDF systems.
- Presented research findings at the IIM ATM 2025 Conference.

Local Deployment and Integration of Open-Source AI Models

- Deployed and evaluated open-source large language models including Gemma and DeepSeek on Ubuntu Linux systems.
- Integrated local inference workflows into Python-based applications and engineering data pipelines.
- Worked with environment management, dependency optimization, and offline AI-assisted workflows.

Technical Skills

Programming: Python, MATLAB (Basic)

Machine Learning & AI: PyTorch, Scikit-Learn, XGBoost, Feature Engineering, Model Evaluation

AI Deployment: Ubuntu Linux, Local LLM Deployment, Gemma, DeepSeek, Git, GitHub

Data Analytics: Pandas, NumPy, Excel, Power BI, Streamlit

Simulation & Modelling: COMSOL Multiphysics, Finite Element Analysis (FEA), Multiphysics Simulation, Shape Memory Alloy Modelling, Thermo-Mechanical Analysis

Materials Characterization: Optical Microscopy, SEM, EDS/EDAX, XRD, Microhardness Testing, Tensile Testing.

Materials Engineering: Advanced Metal Joining Technologies, Manufacturing Processes, Microstructural Analysis, Mechanical Testing

Certifications

- Quantum Computing Workshop – Shaastra IIT Madras
- NPTEL Elite – Solar Photovoltaics: Principles, Technologies and Materials (IIT Kanpur)
- Deloitte Australia Data Analytics Job Simulation (Forage)
- Hewlett Packard Enterprise Software Engineering Job Simulation (Forage)
- TECHNOVATION 2026 and Kurukshetra 2026 Project Presentation Certificates

Languages

English (Professional Working Proficiency – C1)

Tamil(Native)

Hindi (Working Proficiency)

Japanese (Basic Conversational Proficiency)

Technical Interests

Computational Materials Science | Materials Informatics | Finite Element Analysis | Scientific Machine Learning | Advanced Manufacturing | Semiconductor Technologies